A SECURE DIGITAL IDENTITY WALLET FOR SOUTH AFRICA

Flash ID

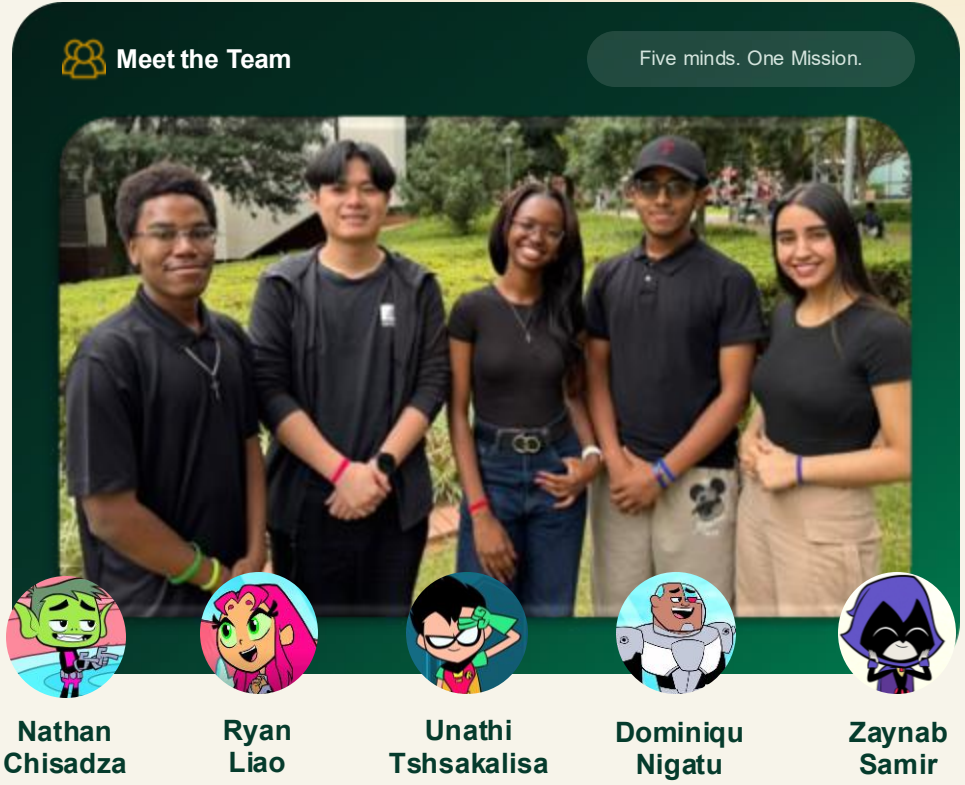
PROVE YOURSELF IN A FLASH.

Your official identity, ready when you are.

Store, share and verify official credentials on web and mobile, so an identity check takes seconds instead of paperwork.

Final Demo

Tech Titans
COS 301 Capstone 2026



WHY FLASHID

Identity in South Africa still runs on paper.

Banks, hospitals, police, DLTCs and Home Affairs all depend on physical documents that can be lost, copied or faked.

THE GOAL

Proof of identity that is instant, private and impossible to fake, even with no signal.

Fragile documents

IDs and licences get lost, stolen or damaged, and replacing them takes weeks.

Copies everywhere

ID copies travel over WhatsApp and email with no control over where they end up.

Hard to trust

A photocopy cannot prove it is real, still valid, or that it belongs to the holder.

No signal, no check

Roadblocks, clinics and rural areas cannot count on a live connection.

HOW IT WORKS

One trusted digital credential. Four simple steps.

1

GOVERNMENT

Issues a verified, digitally signed credential

2

CITIZEN

Stores it in the FlashID wallet on a trusted device

3

CITIZEN

Chooses exactly which fields to share

4

ORGANISATION

Scans the QR and verifies it in seconds

SECURE BY DESIGN

INSTANT VERIFICATION

PRIVACY PROTECTED

BUILT FOR SA

THE ECOSYSTEM

Three audiences. Two apps. One source of truth.

Citizens

Mobile wallet · Web portal

- Wallet & selective-disclosure QR
- Emergency medical profile
- Security alerts & activity

Government admin

Web dashboard

- Register institutions & API keys
- Issue, revoke & reinstate
- POPIA audit log

Officials

Web portal · Mobile verifier

- Onboard citizens from the registry
- Scan & verify, online or offline
- Emergency break-glass access

BUILT WITH Next.js · React Native (Expo) · .NET Clean Architecture · Azure SQL · Cosmos DB · Key Vault · Azure Face Liveness · Government Registry API

SECURITY · LOGIN & SESSIONS

Locking the front door.

Logging in

Auto lockout
5 wrong passwords lock the account for 30 minutes.

BCrypt hashing
Work factor 12. We never store a password.

Password rules
10+ characters with upper, lower and a digit.

Same error message
“Invalid email or password”, so emails aren’t revealed.

New device check
6-digit email code, valid 10 minutes, 5 tries.

Hashed secrets
OTPs and device tokens stored as SHA-256 hashes.

Sessions

HttpOnly cookie
JavaScript can’t read the token

Logout everywhere
a token version bump kills every old session

Expiry
8 hours, or 30 days with remember me

CSRF protection

Cookie `csrf_token` sent automatically

Header `X-CSRF-Token` only our page can set it

Must match, or the request gets 403.

SECURITY · QR & SIGNING

A QR code that can’t be faked, reused or copied.

1 Pick fields
Share only what’s needed

2 Sign in Key Vault
ES256, key never leaves

3 Lives 60 seconds
Unique ID (jti)

4 Official scans
Signature checked by kid

5 Used once
Cosmos DB claim + Active

If any check fails, the scan is rejected and logged. A screenshot or a second scan never works.

Rolling keys

Every 90 days
Key Vault creates a new key version

Daily job
activates it and retires the old one

Retired still verifies
old QRs keep working; revoked keys fail

Offline credentials

SD-JWT
each field salted and hashed separately

Selective disclosure
verifier sees only chosen fields

Bound to the phone
proof re-signed every 5 seconds

Proving who you are

Azure Face liveness
a live person, not a photo

Face match
selfie matched to registry portrait

Registry check
all three must pass to onboard

SECURITY · ABUSE, DATA & ACCESS

Watching, protecting and keeping a trail.

Stopping abuse

Rate limiting
10 logins a minute per IP and 300 a minute overall. Extra requests get 429.

Fraud scoring
Points for impossible travel, new devices and failed logins. High scores raise an alert.

Secure my account
Re-enter your password, then log out other devices or reset it.

Protecting data

SQL injection
EF Core parameters only. No raw SQL.

AES-256-GCM
Emergency fields encrypted at rest.

Certified PDFs
SHA-256 fingerprint catches any edit.

Mobile SecureStore
Keychain / Keystore, this device only.

Access & infrastructure

Role-based access
Citizen, official, government admin.

Registry API key
X-API-KEY, constant-time compare.

HTTPS + HSTS
Every request encrypted.

Safe errors
No stack traces in production.

Audit log
48 event types, reviewed by admins.

FULL SYSTEM

LIVE DEMO

03

Wow factors

Three things a paper ID has never been able to do.



Offline QR

Verify with no signal on either phone.



Emergency QR

Medical info from a locked phone.



Impossible travel

Stops account takeover live.

WOW FACTOR 1 · OFFLINE QR

Verified with zero bars, on both phones.

0

internet connections needed at the moment of verification

SD-JWT · ES256 · device-bound

A copied or filmed code stops verifying within about 90 seconds.

1 Prepared while online

The wallet receives a signed offline package and keeps it encrypted on the phone for up to 30 days.

2 Animated QR, bound to the phone

The phone's device key re-signs the code every 5 seconds, so a screenshot or recording is useless.

3 Verified on the official's phone

Issuer signature, revocation list and device binding are all checked locally.

4 Synced when back online

The result is written to the audit trail exactly once.

WOW FACTOR 2 · EMERGENCY QR

When you can't speak, your phone can.



One tap from the lock screen

A Quick Settings tile shows the code without unlocking the phone.

Signed by the phone's hardware

A StrongBox key signs each code. It is short-lived and single-use.

Officials only, break-glass

A reason is required. Only the medical fields and contacts the citizen chose are shown.

Everyone is told

The citizen and their emergency contacts are emailed, even after offline access.

WOW FACTOR 3 · FRAUD DETECTION

Nobody flies Johannesburg to London in 22 minutes.

Johannesburg, ZA
Sign-in · 10:02
KNOWN

≈ 9,000 km in 22 minutes

London, UK
Sign-in · 10:24
SUSPICIOUS

≈ 24,500
km/h needed

900
km/h limit (faster than a commercial flight)

Instant security alert

The citizen sees the route on a map and why it was flagged.

QR generation frozen

A high-risk event blocks new QR codes for 30 minutes, so a thief cannot present the ID.

Secured in one step

After the citizen re-enters their password, all sessions end, the suspicious device is removed and the block lifts.

FlashID

PROVE YOURSELF IN A FLASH.

Thank you. Questions?

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